

EE 605: Error Correction Codes

Problem Set 1

1. Consider a $(5, 3)$ -linear code over $GF(4)$, defined by the following generator matrix:

$$G = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix}$$

- (a) Find the parity check matrix of this code.
- (b) Characterize the error-correcting and error-detection capabilities of the code.
2. (a) Show that the set of integers under the usual subtraction operation does not form a group.
- (b) Solve the set of equations $2x + y = 3, x + 2y = 3$ over $GF(4)$.
- (c) Consider a set of four elements $\{0, 1, 2, 3\}$. Does this set with the $+$ and $\cdot$ operations defined below constitute a field?

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	2
2	2	3	0	1
3	3	0	1	0

$\cdot$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	3	1
3	0	3	1	2

3. Let p be a prime number and consider the set $\{0, 1, \dots, p-1\}$ with mod- p arithmetic. Show that every non-zero element in the set has a multiplicative inverse.
4. (a) Find the probability of undetected error for the $(8, 4)$ -extended Hamming code and the $(8, 7)$ -simple parity check code, when used over the $BSC(p)$ channel. Compare the two quantities for $p = 1/2$.
- (b) Consider the $(5, 1)$ -binary repetition code. List the vectors which will be decoded to (00000) under nearest neighbor decoding. Also, assuming both messages to be equally likely, what will be the probability of error if this code is used over a $BSC(p)$ channel?
5. Consider a $(5, 3)$ -binary code C with generator matrix as follows:

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

- (a) Convert G to systematic form.
- (b) Find the parity check matrix of the code C .
- (c) List all the codewords of the dual code of C .

6. Consider a $(6, 3)$ -binary code C with parity check matrix as follows:

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

- (a) Construct the coset decomposition of this code, choosing the lowest weight available vector as each coset leader.
 - (b) Construct the syndrome decoding table.
 - (c) Use syndrome decoding to decode the received vectors $r_1 = (000011)$ and $r_2 = (111100)$.
7. Recall that each element h in a finite group G is associated with a cyclic subgroup. We will say the group G is cyclic if it is equal to the cyclic subgroup generated by one of its elements g , so that $G = \{e, g, g^2, \dots, g^{n-1}\}$. In other words, there is an element with order n , where n is the size of the group G .

- (a) Show that any cyclic group G is abelian.
- (b) Show that a group of size 3 will be cyclic.
- (c) Is the same true for groups of size 4?
- (d) Is the group $\{0, 1, \dots, 7\}$ with mod-8 arithmetic a cyclic group? If yes, identify all elements whose cyclic subgroup generates the group. *Remark: Other than doing this in a brute-force way by enumerating all the cyclic subgroups, you might want to try to show that the order of element g^i in a cyclic subgroup generated by g is $n/\gcd(i, n)$.*

8. *Cyclic groups.*

- (a) Consider the multiplicative group of integers modulo 5 given by $G = \{1, 2, 3, 4\}$ under mod-5 multiplication. Find the order of the cyclic subgroup generated by each element in G . Is G a cyclic group? Justify your answer.
- (b) Let $p > 3$ be prime, Consider the multiplicative group of integers modulo p given by $G = \{1, 2, \dots, p-1\}$ under mod- p multiplication. This group is known to be cyclic. Show that not all non-unity elements can generate G as their cyclic subgroup.
- (c) Instead, now consider $n = 4$, and $G = \{1, 2, 3\}$ under the multiplication rule given below. Show that the above claim is not true in this case, i.e., all non-unity elements do generate G .

.	1	2	3
1	1	2	3
2	2	3	1
3	3	1	2

9. Show that for an (n, M, d) -code, the error correction capability of any decoder cannot be larger than $\lfloor \frac{d-1}{2} \rfloor$, i.e., there is no decoder which can correct all error patterns with Hamming weight upto $\lfloor \frac{d+1}{2} \rfloor$.

10. Consider one out of M codewords, chosen uniformly at random, being transmitted over a BSC(p) with $p < 1/2$. At the receiver, show that Maximum Likelihood Decoding = Maximum A posteriori Decoding = Nearest Neighbour Decoding.

Maximum Likelihood Decoding: We find the codeword X that gives the maximum-likelihood $P(Y|X)$ for the given received word Y

Maximum A posteriori Decoding: We find the codeword X that gives the maximum-a posteriori probability $P(X|Y)$ for the given received word Y

Nearest Distance Decoding: We find the codeword X that has least Hamming distance with the received word Y

11. (a) Consider the multiplicative group of integers modulo 7, given by $G = \{1, 2, 3, 4, 5, 6\}$ under modulo 7 multiplication. Find the order of the cyclic subgroup generated by each element in G . Is G a cyclic group? Justify your answer.
- (b) Let $p > 3$ be prime. Consider the multiplicative group of integers modulo p , given by $G = \{1, 2, \dots, p - 1\}$ under mod- p multiplication. This group is known to be cyclic. Show that not all non-unity elements can generate G as their cyclic subgroup.
12. Find a basis of the null space of the subspace spanned by $\{01110, 01001, 11100\}$.